P A L O M A R O B S E R V A T O R Y · C A L T E C H

FACSUM

Web Interface — User Manual

Telescope	P200 Hale, Palomar Mountain
Platform	Browser-based (PHP / JavaScript)
Revision	1.2 · August 2026
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1 Introduction

1.1 Overview

FACSUM (Facility Summary) is a browser-based observing tool for the Palomar Observatory 200-inch Hale Telescope (P200). It brings four key observing-support functions together in a single page:

- Live monitoring of P200 Telescope Control System (TCS) telemetry — apparent pointing, time, guide offsets, tracking rates, Cassegrain ring angle and focus
- Loading, editing and managing observer target lists, with real-time airmass computation for each target
- A live sky display showing the azimuthal projection of the sky, telescope pointing, solar-system bodies, and the all-sky camera feed
- An airmass plot showing when each target is best observable across the night

FACSUM runs directly on **Observer1** (`observer1.palomar.caltech.edu`), where a PHP built-in web server listens on `127.0.0.1:18080`. It is used from the **Portal** workstation. The startup script `start_facsum.sh` (run on Portal) establishes an SSH tunnel to Observer1, starts the web server if it is not already running, and opens FACSUM in a Firefox window — no manual browser navigation is required. FACSUM is not directly reachable from outside the Palomar control room network.

Because FACSUM now runs *on* Observer1, target lists are read and written through ordinary filesystem calls. There is no SFTP step and no login prompt inside the browser: the only password you may be asked for is the Observer1 SSH password requested by `start_facsum.sh` at launch, and only when a healthy session is not already open.

> **New in Revision 1.2:** interface zoom control (§1.3); collapsible *Parsing Options* (§4.2); double-click cell editing in the Targets Table (§4.4); SIMBAD name resolution in the Add Single Target form (§4.5); all-sky camera alignment controls (§5.3); an *All / Selected* target display toggle (§5.5); the target file browser is now a local Observer1 browser rather than an SFTP client (§7); NGPS format support and decimal-degree coordinate handling (Appendix A).

1.2 Interface Layout



Figure 1 — FACSUM interface with a live TCS connection and a loaded target list. Left column: Telemetry (top) and Target Lists (bottom). Right column: Sky Display (top) with all-sky camera overlay, target dots, and planetary markers, and Airmass Plot (bottom).

The interface is divided into four panels arranged in two columns. Each panel has a coloured accent to help orient quickly:

Panel	Position	Accent	Purpose
P200 Telemetry	Left, top	Cyan	Live TCS data: apparent pointing, time, offsets, rates, Cass ring and focus
Target Lists	Left, bottom	Orange	Load, edit, save, and send targets to the telescope
Sky Display	Right, top	Purple	Real-time sky projection with all-sky camera overlay
Airmass Plot	Right, bottom	Purple	Target observability curve across the night

Resizing panels: Three drag handles let you redistribute screen space to suit your workflow. Drag the *vertical bar* between the two columns to adjust their widths. Drag either *horizontal bar* — one between Telemetry and Target Lists, one between the Sky Display and the Airmass Plot — to adjust how space is split between the panels above and below it.

The **Airmass Plot** section can be collapsed by clicking its header bar, freeing additional vertical space when not needed. Its resize handle disappears while collapsed.

1.3 Interface Zoom

The Telemetry panel header contains a zoom control — –, a percentage readout, and **+** — which scales the entire FACSUM page. This is useful on the control-room displays, where the default browser text size can be

uncomfortably small at a distance.

- Range **50 % to 200 %**, in 10 % steps.
- The chosen level is remembered in the browser and is reapplied on the next page load.
- On a browser profile that has never run FACSUM, the interface starts at **150 %**.

The panel drag handles and the input-validation tooltips compensate for the zoom factor, so resizing behaves identically at any zoom level. One exception: dragging to pan the sky canvas (§5.6) moves the sky by the zoom factor, so at 150 % the view travels about 1.5× as far as the cursor.

2 Quick Start

The following steps describe the typical workflow at the start of a night run.

1. **Launch FACSUM.** On the **Portal** workstation, run `start_facsum.sh`. The script reuses a healthy Observer1 session if one exists; otherwise it prompts once for the Observer1 password, opens the SSH tunnel, starts the web server, and opens FACSUM in Firefox. The interface loads and connects to the TCS automatically — live telemetry begins updating within a few seconds.
2. **Verify the connection.** Confirm the green *Connected* indicator on the right of the Telemetry panel header. If it shows *Disconnected*, click **Connect** to re-establish.
3. **Load your target list.** Click **+ Load Target List** in the Target Lists panel header — or the large **+ Load Target List** button shown in the middle of the empty table. This opens the file browser (§7), where you can navigate the Observer1 target directory and open your file. FACSUM parses it and populates the table automatically. See §4.1 for alternatives.
4. **Check target observability.** The Targets Table shows a live, colour-coded airmass for each object. Sort the table by **Airmass** to find the best target for right now.
5. **Select a target.** Click any table row to select it. The selected target is highlighted on the sky display with a crosshair, and its airmass curve is drawn in the plot below.
6. **Send the target to the telescope.** With a row selected, click **Load to Telescope**. This sends a `COORDS` command to the TCS, loading the target's coordinates.

> **Important — COORDS does not slew:** Sending a target with *Load to Telescope* loads coordinates into the TCS but does not move the telescope. Initiate the slew separately in the TCS software (e.g. TCSNG) before observing.

3 Telemetry Panel



Figure 2 — Telemetry panel in the connected state. The orange Loaded block shows the last target sent to the TCS; the cyan Current block shows the live telescope pointing. The time strip displays UT, LST, HA, and live airmass.

3.1 Connecting to the TCS

FACSUM connects to the TCS automatically about half a second after the page loads. A background daemon on Observer1 (`tcs_daemon.py`) holds the TCP connection to the TCS, polls it four times a second, and writes parsed telemetry to a small JSON file; the browser reads that file roughly ten times a second and redraws only when the daemon has published new data.

If the connection is lost or was manually stopped, click **Connect** in the panel header to restart the daemon and resume polling. A status LED and label on the *right side* of the header reflect the current connection state, colour-coded to match:



Connected state *Disconnected state*

LED colour	Label	Meaning
● **Green** (slow pulse)	Connected	Polling the TCS successfully; all fields update continuously
● **Yellow** (fast pulse)	Reconnecting…	The browser lost contact with the Observer1 server and is retrying (the button reads *Connecting…*)
● **Yellow** (fast pulse)	TCS Reconnecting…	The daemon is running but has lost its link to the TCS and is retrying
● **Red** (static)	Disconnected	No TCS connection — the daemon is idle, stopped, or has published nothing yet
● **Red** (static)	Daemon stopped	The daemon shut down cleanly, e.g. after **Disconnect** or a heartbeat timeout
● **Red** (static)	Error	The daemon could not reach the TCS; the underlying message appears in the status bar

The LED pulses slowly while connected and faster while connecting, providing a passive at-a-glance indication that the link is live. It is static when disconnected. To the left of the LED, a small *updated HH:MM:SS UTC* stamp shows when the daemon last published a telemetry frame, and a status bar along the bottom of the panel carries any message the daemon reports.

Click **Disconnect** to stop the daemon cleanly and release the TCS connection. Closing or navigating away from the tab also sends a disconnect request automatically.

> **Note:** Keep only one browser tab connected at a time. A second tab opening a connection will conflict with the daemon already started by the first. Switching to another application (TCS, a terminal) is safe — FACSUM deliberately keeps its connection alive when its tab is not visible.

3.2 Connection Resilience

Brief network hiccups between Observer1 and the TCS are common and are handled without operator intervention. Two mechanisms are worth knowing about, because they explain why the display sometimes holds still instead of alarming.

Mechanism	Behaviour
TCS grace period	If the daemon reports that it has lost the TCS, the displayed values are frozen and no warning is shown for the first **12 seconds**. The daemon normally reconnects within a couple of seconds. Only if the outage outlasts the grace period does the header switch to **TCS Reconnecting…**, and even then the last-known field values stay on screen rather than blanking.
Browser auto-reconnect	If the browser cannot reach the Observer1 web server for about **5 seconds** of continuous failures, FACSUM stops polling and retries the connection up to **5 times, 5 seconds apart**. Progress is reported in the status bar (*"Connection lost — reconnecting (attempt 2 of 5)…"*). If all attempts fail, the panel returns to *Disconnected* and the status bar invites you to click **Connect**.
Heartbeat	Every telemetry poll renews a heartbeat file. If the heartbeat goes stale for **120 seconds** — for example because the browser was closed without a clean disconnect — the daemon drops its TCS connection on its own and waits idle until a browser returns. This prevents an abandoned session from holding the TCS link open.

3.3 Position & Time Fields

The Telemetry panel shows pointing information in two side-by-side blocks, followed by a hardware strip and a time strip. The **Loaded** block changes only in response to your actions in the Target Lists panel. The **Current** block reflects live TCS data and updates continuously while connected.

Loaded block — last target sent to the TCS

Populated when **Load to Telescope** succeeds (§4.7). All four fields show "—" until the first successful send in the current session.

> **When the block resets:** the Loaded block is cleared back to "—" whenever the list is re-parsed — loading a list, pressing *Parse Text*, or adding or deleting a target — even if a target was sent earlier in the session. After an in-place cell edit (§4.4) it shows the *currently selected* row rather than the last target actually sent. Treat the **Current** block and the TCS console as authoritative for what the telescope is holding.

Field	Description
LOADED TARGET	Name of the last target sent to the TCS (truncated to 12 characters if longer)
LOADED RA	Right Ascension of that target (HH:MM:SS.s)
LOADED DEC	Declination of that target (±DD:MM:SS)

Field	Description
LOADED EQUINOX	Coordinate epoch of that target, normalised for display (e.g. `J2000.0`; epochs before 1984 are shown with a `B` prefix)

Current block — live telescope pointing

Field	Description
CURRENT TARGET	Target name currently set in the TCS (truncated to 12 characters if longer)
CURRENT RA	*Apparent* Right Ascension of the telescope, computed as LST − HA from TCS encoder data (HH:MM:SS.ss)
CURRENT DEC	*Apparent* Declination, derived from the reported telescope azimuth and elevation together with the Palomar latitude (±DD:MM:SS.s)
CURRENT EQUINOX	Coordinate epoch of the current pointing — sourced from the last target sent by FACSUM, as the TCS does not report this value in its telemetry stream. Reads "—" until the first successful send.

> **Why apparent coordinates?** The TCS reports the catalog (J2000) position it was given, which drifts away from where the telescope is actually pointing once offsets and non-sidereal rates are applied. FACSUM instead derives the pointing from the encoders, so the Current block matches the TCS console display rather than the original catalog entry.

Hardware strip

Field	Description
RA OFFSET	Current RA guide/offset correction, suffixed **E** or **W**. Shown in arcseconds, or in degrees once the magnitude reaches 1000″.
DEC OFFSET	Current Dec guide/offset correction, suffixed **N** or **S**, with the same unit switch.
RA RATE	Non-sidereal tracking rate in RA (arcsec/hr); reads `0.0"/hr` when not tracking
DEC RATE	Non-sidereal tracking rate in Dec (arcsec/hr)
CASS RING	Cassegrain ring rotation angle, normalised to 0–360°
FOCUS	Secondary mirror position (mm)

Time strip

Field	Description
UT	Universal Time as reported by the TCS (HH:MM:SS)
LST	Local Sidereal Time at Palomar (HH:MM:SS)
HA	Hour Angle, prefixed E or W of meridian
AIRMASS	Airmass at the current telescope pointing, as reported by the TCS

Note: LST from the telemetry drives almost everything else on the page — the airmass column in the target table (§4.3), target and solar-system positions on the sky display (§5), and the telescope crosshair. If the TCS is disconnected, airmass cells show "—" and no targets or bodies are plotted on the sky. The UT field additionally sets the *NOW* marker in the airmass plot, so that marker follows observatory time rather than the workstation clock.

4 Target Lists Panel

4.1 Loading a Target File

There are three ways to get targets into FACSUM:

- **From Observer1 (recommended):** Click **+ Load Target List** in the panel header. This opens the file browser (§7), which lists the Observer1 target directory. Navigate to your file and click **Open** (or double-click it). FACSUM reads and parses the file automatically. When the table is empty, a second, larger **+ Load Target List** button appears in the middle of the pane and does the same thing.
- **By pasting text:** Click the **Edit Text** tab, paste or type your target list, and click ■ **Parse Text**. This is useful for quickly trying a short list without saving a file.
- **One target at a time:** Click **+ Add Single Target** at the bottom of the panel and fill in the form — optionally letting SIMBAD resolve the coordinates from the object name (§4.5).

The filename of the currently loaded list is always shown in the upper-right corner of the panel header (e.g., **targets123.csv**); it reads — when no list is loaded. Opening a file — or adding, editing or deleting a target — activates the **Save Target List** and ✕ **Clear List** buttons.

> **Note:** pasting text and pressing ■ **Parse Text** populates the table but does not by itself activate *Save Target List*. Adding or editing any target afterwards activates it, and the save then writes the full text including your pasted lines.

FACSUM supports three file formats — **CSV**, **PTIC**, and **NGPS** — and auto-detects the format of any file you open. See Appendix A for the full format specification.

4.2 Parsing Options

Below the tab bar is a collapsible **Parsing Options** section. Click the summary line to expand or collapse it. In most cases the defaults work without adjustment, and the section can be left closed.

Option	Choices	Notes
File Type	Auto-detect · CSV · PTIC	Auto-detect scores every format/delimiter combination against the file and picks the best match, showing a summary such as *"auto-detected: CSV comma (12/12 lines)"*. Override only if auto-detection picks the wrong format. NGPS is recognised from its header row and does not need (or have) a radio button.
Delimiter	Auto · Comma · Semicolon · Tab · Space · Other	Applies to CSV files only. *Auto* examines up to ten sample lines and picks the most frequent separator among comma, semicolon, tab and pipe. Choose *Other* and type a character (e.g. |) for anything else; typing in that box selects the *Other* radio for you.
Rate Units	Proper Motion (0.0001s/yr, 0.001"/yr) Non-sid Tracking (arcsec/hr, arcsec/hr) Non-sid Tracking (sec/hr, arcsec/hr)	Tells FACSUM how to interpret the rate columns of your file, and therefore what motion flag to send to the TCS. *Non-sid Tracking (arcsec/hr, arcsec/hr)* is the default and covers most solar-system target lists. The table's rate column headers update to match the chosen unit mode.

Option	Choices	Notes
Apply Rates	Checkbox (checked by default)	When checked, non-sidereal tracking rates are included in the `COORDS` command sent to the TCS. Uncheck for sidereal targets whose file rates should be ignored.

> **Format follows the row:** a single file may mix formats, and FACSUM parses each line independently. When you select a row, the File Type / Delimiter controls and the detection label update to reflect the format of *that* line.

4.3 Targets Table



*Figure 3 — Target Lists panel with WIRC_Photometric_Standards.csv loaded (filename shown in the header). S754-C is selected (orange highlight) and marked as observed (✓). The **Load to Telescope** button is active. The bottom bar shows the Add, Delete, and Clear List controls.*

Column	Description
✓ *(Observed)*	Only visible when the table holds at least one target. See §4.8.
Name	Object name from the target file
RA	Right Ascension (HH:MM:SS.s). Shown in *italics* if FACSUM converted it from a decimal value — hover for the original.
Dec	Declination (±DD:MM:SS), with the same italic/hover convention
Epoch	Coordinate epoch (e.g. 2000)

Column	Description
Airmass	Current airmass, computed from LST telemetry and updated continuously, colour-coded **green** <1.5, **yellow** 1.5–2.0, **orange** 2.0–3.0, **red** >3.0. Shows *Down* when the object is at or below the P200 altitude limit (6.5°, airmass ≈ 8.8), and "—" when LST is unavailable. This column is computed and cannot be edited.
PM RA / Rate RA	Proper motion or tracking rate in RA. The header label changes with the Rate Units setting — `PM RA (0.0001s/yr)`, `Rate RA (arcsec/hr)` or `Rate RA (s/hr)`.
PM Dec / Rate Dec	Proper motion or tracking rate in Dec.
Notes	Any trailing text from the source line (text after a `!` character, plus unrecognised trailing tokens)

NGPS files add two further columns, **Priority** and **Mag / Filter**, and rename the rate columns to **Offset RA** / **Offset Dec** and Notes to **Comment** (Appendix A.3).

Lines that fail to parse

Any line FACSUM cannot interpret is listed in red at the top of the table as *"Failed to Parse: …"*, with the raw text shown. These rows are informational only — they cannot be selected or sent. Fix them in the **Edit Text** tab (§4.6), where the same lines are highlighted red as you type.

Sorting

Click any column header to sort by that column — click once for ascending, again for descending, and a third time to restore the original file order. An arrow (▲ / ▼) marks the active sort column. Sorting by **Airmass** is a quick way to find the most observable target right now. Sorting is reset whenever the list is re-parsed.

Selecting a target

Click a row to select it (it highlights in orange). Click it again to deselect. Only one target can be selected at a time. Selecting a target:

- Draws a larger dot and crosshair for that target on the sky display
- Draws that target's airmass curve in the Airmass Plot, and puts its name in the plot's header bar
- Enables the **Load to Telescope** button (§4.7) and – **Delete Selected**
- Updates the Parsing Options controls to reflect that line's own format

4.4 Editing the Table Directly

Double-click any editable cell to change its value in place, without switching to the Edit Text tab. The cell becomes a text field; press **Enter** to commit, **Escape** to cancel. Clicking elsewhere also commits. The Airmass column is computed and the ✓ column is a toggle, so neither is editable.

Values are validated as you type, and an invalid value *refuses to commit* — the field stays open with a red outline and an explanatory tooltip:

Field	Accepted	Example message
Name	Any non-empty string; at most 17 characters on a PTIC-format line	*Name too long for PTIC format (max 17 chars)*
RA	`HH:MM:SS[.ss]`, hours 0–23	*Impossible RA — minutes must be 0–59*

Field	Accepted	Example message
Dec	`±DD:MM:SS[.s]`, magnitude ≤ 90°	*Invalid syntax — expected ±DD:MM:SS.s*
Epoch	`2000`, `2000.0`, `J2000`, `B1950`	*Invalid epoch — expected e.g. 2000, J2000, B1950*
Rates	A signed number	*Invalid rate — expected a numeric value*
Notes, Priority, Mag / Filter	Free text	—

On commit, FACSUM rewrites only the edited line in the raw text — in that line's own original format (Appendix A.4) — and then re-parses the whole list, so the table and the text you would save can never drift apart. Your sort order and row selection are preserved across the edit.

4.5 Adding, Deleting & Clearing Targets

Adding a single target manually

Click **+ Add Single Target** at the bottom of the panel to open a data-entry form. **Name**, **RA** and **Dec** are required — the **Add Target** button stays disabled until all three are valid — and fields are validated exactly as in §4.4, with a green outline for good input and a red outline plus tooltip for bad. The form separates *Non-sidereal Rates* (arcsec/hr) from *Proper Motion* (mas/yr), and has a free-text Notes field. Pressing **Enter** anywhere in the form submits it when the required fields are valid.

The new target is appended to the list as a comma-delimited line and is immediately available for selection and sending. This is the quickest way to add a target of opportunity.

SIMBAD name resolution

Type an object name and click the ■ button beside it to look the object up in **SIMBAD** (via the CDS Sesame name resolver, proxied through Observer1). The form is replaced by a result card:

- **Object matched** — shows the resolved Name, RA, Dec, Epoch (J2000) and proper motions in mas/yr. Click **Add Target ✓** to add it directly, or ← **Edit** to copy the values back into the form for adjustment before adding.
- **Not found** — shows a short explanation (*Target not found*, *Could not reach SIMBAD*, *Server error*) with an expandable *Details* section carrying the technical response. Click ← **Back** to return to the form.

> **Note:** the lookup requires outbound network access from Observer1 to `cdsweb.u-strasbg.fr` and times out after 15 seconds. It resolves coordinates and proper motions only — never rates, magnitudes or exposure times.

Deleting a target

Select a row, then click − **Delete Selected** to remove it. The corresponding line is deleted from the raw text and the list is re-parsed. The deletion cannot be undone from the table view; use the Edit Text tab (§4.6) and its Undo button if needed.

Clearing the target list

Click ✕ **Clear List** (separated by a | from the Delete button) to unload everything. This removes all rows, clears the Edit Text area and the detected-format label, hides the Observed column, resets the filename display to —, disables Save and Load to Telescope, and resets the file browser back to `/observer/observer/targets` so the next **Load Target List** opens there. The sky display and airmass plot clear immediately. This action cannot be undone.

4.6 Edit Text Tab

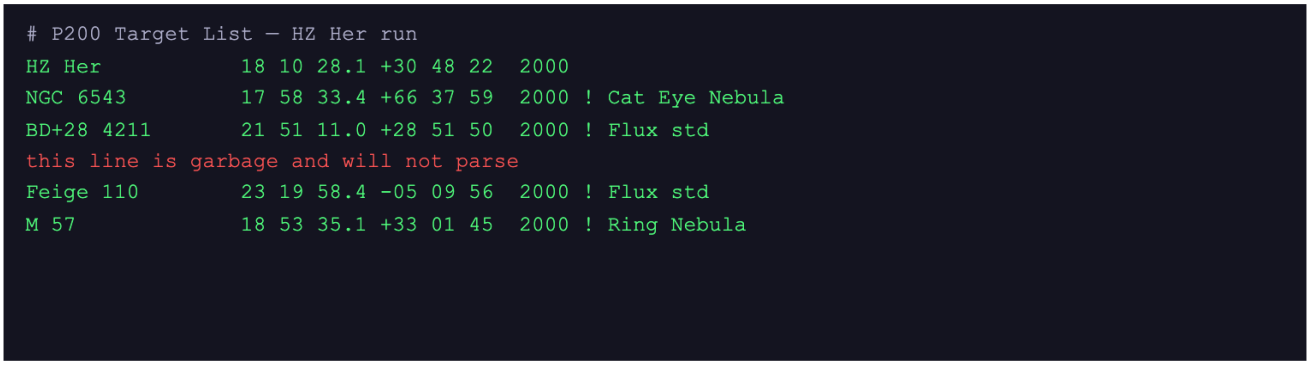


Figure 4 — Edit Text tab. Lines are coloured in real time: valid targets in green, unparseable lines in red, and comments/blanks dimmed.

The **Edit Text** tab shows the raw text of the loaded target list and lets you edit it directly. It is the single source of truth: everything the table shows is parsed from this text, and it is exactly what gets written when you save. A live syntax-highlighting overlay colours each line as you type:

- **Green** — line was successfully parsed as a target
- **Red** — line could not be parsed (check format or delimiter settings)
- *Dimmed* — comment line (starts with `#` or `!`), blank line, or an NGPS header row

Three action buttons appear at the bottom of the Edit Text pane:

- ■ **Undo** / ■ **Redo** — step backward or forward through the editing history (equivalent to Ctrl+Z / Ctrl+Y)
- ■ **Parse Text** — apply the current text and switch back to the table view

Switching to the **Targets Table** tab also re-parses the text, so you can move between the two views freely.

4.7 Sending a Target to the Telescope

The **Load to Telescope** button sits in its own bar between the Parsing Options and the table. It is always visible but stays disabled until a table row is selected. Clicking it sends a COORDS command to the P200 TCS over a direct network connection (see Appendix B for the command format).

The button shows a brief status as the command is processed:

- **Sending…** — command is in flight; button is temporarily disabled
- **Target Loaded!** — TCS acknowledged the coordinates; the **Loaded** block in the Telemetry panel updates, and a ✓ is placed in the target's Observed column (§4.8). Reverts to normal after 2 s.
- **Error** — command failed; check the network connection or the server log (reverts after 2.5 s). The detailed error is written to the browser console.

If the selected row has an RA or Dec that cannot be interpreted, FACSUM refuses to send and shows an alert instead.

Two things to check before sending:

- Rates are only included when *Apply Rates* is checked, *Rate Units* is one of the two Non-sid Tracking modes, **and** at least one of the target's two rate values is non-zero. Otherwise zero rates are sent regardless of what the file contains.
- COORDS loads coordinates but **does not slew** the telescope. Issue a separate slew command in the TCS software after sending.

4.8 Observed Column

When the table holds at least one target, its first column shows a ✓ for each target that has been observed during the session. The column is hidden when the table is empty.

- **Automatic marking:** when a target is successfully sent to the telescope via **Load to Telescope** and the TCS acknowledges the command, its row is marked automatically.
- **Manual toggle:** click the ✓ cell of any row to add or remove the mark by hand — useful for correcting the record or marking targets observed by other means.
- **Row highlight:** observed rows are tinted faint orange to make them visually distinct from unobserved targets at a glance.
- **Session only:** the observed state is held in memory and is never written to the target file. It survives sorting, but is cleared whenever the list is re-parsed — loading a list, parsing edited text, editing a cell, or adding or deleting a target.

4.9 Saving a Target List

Click **Save Target List** in the panel header to open the file browser in save mode (§7). A *Filename* field appears, pre-filled with the name of the currently loaded list, and the action button reads **Save Here**. Navigate to the directory you want, adjust the filename, and click **Save Here** to write the current text — including any edits, additions and deletions — either over the original file or to a new one. Clicking an existing file copies its name into the field, so overwriting is deliberate rather than accidental.

Note: saves are confined to the Observer1 target directory tree. Some older target lists are owned by stale accounts and cannot be written in place; in that case FACSUM replaces the file rather than failing, which also transfers ownership going forward.

5 Sky Display

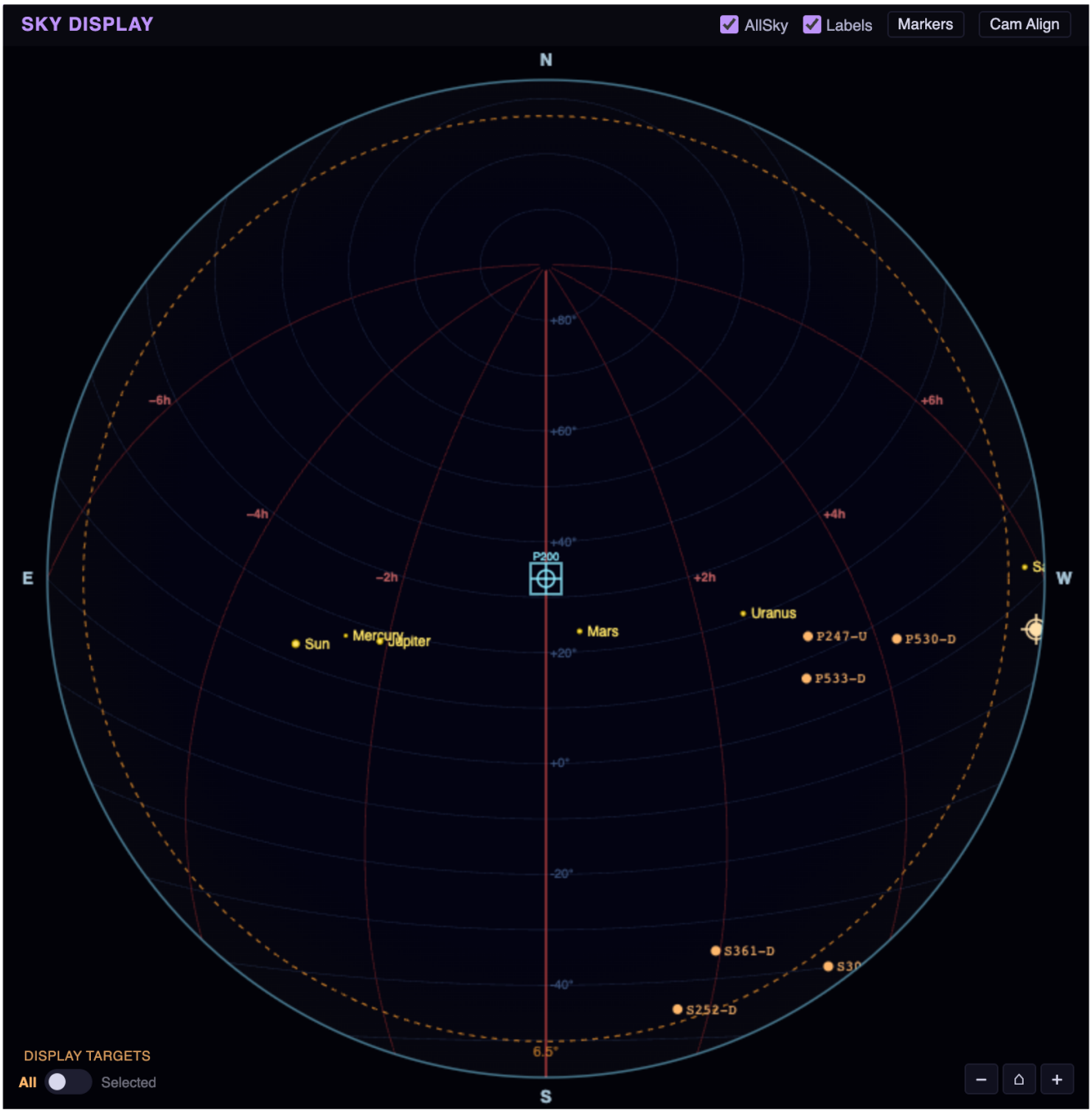


Figure 5 — Sky Display with the all-sky camera overlay and target list active. Orange dots with labels show the WIRC photometric standards. The cyan P200 fiducial marks the current telescope pointing. Yellow markers show solar system bodies. The dashed orange ring is the 6.5° altitude limit; the heavier red line is the meridian.

The panel header carries four controls: the **AllSky** and **Labels** checkboxes, and the **Markers** and **Cam Align** dropdowns. Clicking anywhere outside an open dropdown closes it.

5.1 Sky Projection & Grid

The sky canvas uses an **azimuthal equidistant projection** looking straight up from Palomar Mountain. The zenith is at the centre; objects closer to the horizon appear closer to the edge of the circle.

Orientation follows standard astronomical convention: **North is up, East is to the left**. The grid is fixed in hour-angle/declination space, so an object transiting the meridian appears on the meridian line, and objects at large hour angles appear toward the east or west edges.

Element	Appearance	Meaning
Declination circles	Faint blue rings every 10°	Lines of constant declination; labelled every 20° where they cross the meridian
Hour angle lines & Meridian	**Red** lines at ±2 h, ±4 h, ±6 h; the meridian (HA = 0 h) drawn thicker and brighter	Lines of constant HA, labelled at their Dec +30° crossing (west of the meridian positive); objects cross the meridian at their highest elevation
Altitude limit	**Orange** dashed ring, labelled *6.5°*	The P200 elevation limit — avoid pointing inside this ring
Horizon circle	Cyan outer ring	The horizon (altitude = 0°)
Cardinal labels	N, E, S, W outside the ring	Compass directions
Telescope crosshair	**Cyan** circle + lines + square, labelled *P200*	Current telescope pointing from live TCS telemetry. Always rendered on top of target dots and body labels.
Target dots	**Orange** dots with name labels	Parsed targets above the horizon; the selected target has a larger, paler dot plus a crosshair

Unchecking **Labels** hides the declination and hour-angle values, target names, and body names, leaving the geometry uncluttered. The cardinal letters and the *6.5°* ring label are always drawn.

> **Note:** targets, solar-system bodies and the telescope crosshair all require LST from the telemetry stream. With the TCS disconnected the grid is still drawn, but nothing is plotted on it.

5.2 All-Sky Camera Overlay

Check **AllSky** in the panel header to overlay the live Palomar all-sky camera image on the sky projection. The image is fetched every 30 seconds through a server-side proxy on Observer1 (which caches it for 30 seconds so many browsers can share one camera fetch) and composited at 55 % opacity.

The overlay is not simply pasted on: FACSUM remaps the fisheye image pixel by pixel, using the lens calibration from the original FACSUM `telescope.cfg`, so that features in the camera image land on their true altitude and azimuth in the equidistant grid. If the browser cannot read the image pixels, it silently falls back to a plain crop-and-rotate, which is close but not geometrically exact.

If the camera is unreachable (daytime, dome closed, network outage), the most recent successful image is retained and served until a new one arrives.

5.3 Camera Alignment (Cam Align)

Click **Cam Align** in the panel header for two calibration fields that fine-tune how the all-sky image sits on the grid. Adjust them while watching a recognisable feature — the Moon, a bright planet, or the dome slit — and the overlay redraws live as you type.

Field	Default	Range	Effect
Rotation ° CW	–13	–180 to 180, 0.5° steps	Rotates the camera image clockwise about the zenith to align its north with the grid's north
Scale	1.0	0.5 to 2.0, 0.005 steps	Scales the camera field of view. Values below 1 zoom the camera image in, pushing its horizon beyond the canvas edge.

Both values are stored in the browser, so an alignment you establish once survives page reloads on that workstation. They affect only the overlay — never the projection, the grid, or any computed position.

5.4 Markers (Moon, Sun & Planets)

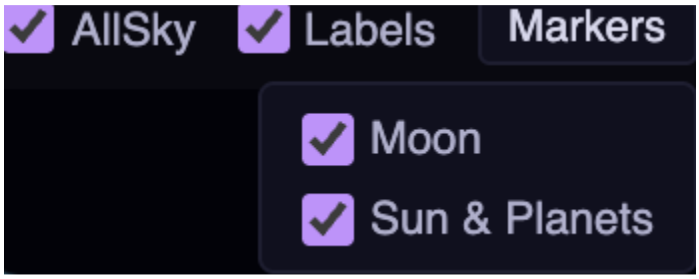


Figure 6 — The Markers dropdown in the Sky Display header.

Click **Markers** in the panel header to open a small dropdown with two checkboxes, both enabled by default:

- **Moon** — draws a faint disc outline at the computed lunar position with the illuminated limb traced on top, so the marker reflects the current phase (crescent, quarter, gibbous, full). The phase comes from the Sun–Moon elongation and the position from the Meeus simplified lunar theory (accuracy ≈ ±1°). This checkbox also controls the Moon's airmass curve in the plot below (§6.1).
- **Sun & Planets** — draws yellow dots for the Sun, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune, with marker radii scaled roughly by apparent brightness and size. Positions come from Keplerian elements (Meeus Table 31.a, accuracy ≈ ±1–2°). All calculations are done in the browser from the current time; no external ephemeris service is needed.

Bodies below the horizon are not drawn. Toggle **Labels** in the header to show or hide body name labels next to each marker.

5.5 Display Targets — All / Selected

A small **Display Targets** switch in the lower-left corner of the sky canvas controls how much of the target list is plotted:

- **All** (default) — every target above the horizon is drawn, each with its name label.
- **Selected** — only the currently selected target is drawn.

Click the switch, or either of the two labels, to change modes. *Selected* is the practical choice with long lists, where dozens of labelled dots can bury the telescope crosshair and the all-sky image.

5.6 Zoom & Pan

Three buttons in the lower-right corner of the sky canvas control the view level:

Button	Action
– (Zoom out)	Shrinks the sky circle within the pane (÷ 1.6 per click, no smaller than the full view)
■ (Reset)	Returns to the default full view with no zoom or pan offset
+ (Zoom in)	Enlarges the sky circle within the pane (× 1.6 per click, up to 20×)

When zoomed in, **click and drag** on the canvas to pan; panning is clamped so the sky circle cannot be dragged entirely out of view. The cursor changes to a grab icon when panning is available. The entire sky circle — grid, markers, cardinal labels, and all-sky overlay — scales and pans together uniformly.

6 Airmass Plot

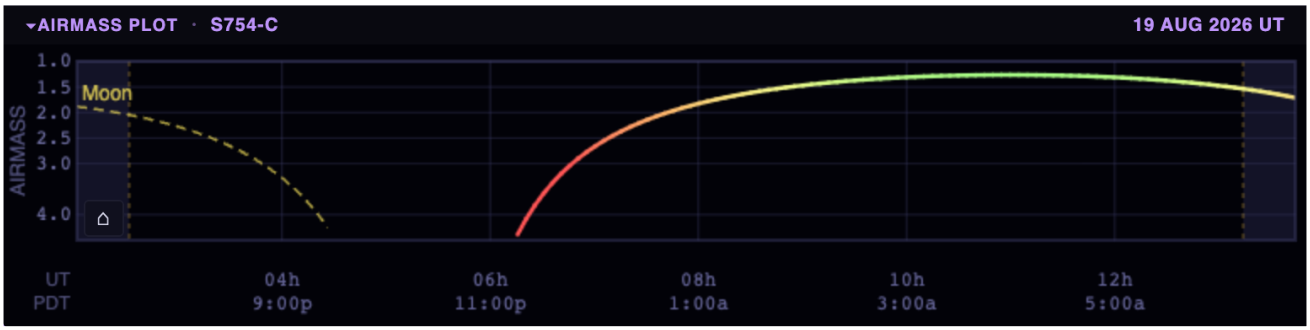


Figure 7 — Airmass Plot showing the selected target S754-C (solid curve, colour-graded green→yellow→orange→red by airmass) alongside the Moon's curve (yellow dashed). The x-axis spans the observing night in UT (top row) and Pacific Time (bottom row).

The Airmass Plot (below the sky canvas, collapsible) shows the observability of the selected target across the entire observing night. It updates as new telemetry arrives and is redrawn at least once a minute so the *NOW* marker stays current.

The section header doubles as a legend: it shows the name of the selected target and the **UT date of the night** being displayed (the date on which sunset falls), e.g. *17 AUG 2026 UT*.

> **Night selection:** FACSUM determines which night to display automatically. Before sunrise it shows the night in progress (last sunset through this morning's sunrise); after sunrise it switches to the upcoming night (today's sunset through tomorrow's sunrise). The plot always extends 30 minutes beyond sunset and sunrise at each end.

6.1 Reading the Plot

Element	Description
X-axis	UT hours across the night, gridded every 2 hours (top label row), with Pacific Time equivalents below (labelled PDT or PST as appropriate)
Y-axis	Airmass from 1.0 (zenith) to 4.5, gridded at 1.0, 1.5, 2.0, 2.5, 3.0 and 4.0. Lower is better.
Dark blue shading	The 30-minute margins before sunset and after sunrise — daylight, not observing time
Amber dashed lines	Exact sunset and sunrise times
Cyan dashed line	Current time, labelled **NOW** in the margin just below the plot. Derived from the TCS UT field, so it reflects observatory time rather than the workstation clock.
Yellow dashed curve	The Moon's airmass across the night, labelled *Moon*. Shown whenever the **Moon** marker is enabled in the Sky Display's Markers dropdown.
Colour-graded target curve	Target airmass, graded continuously by value: **green** at 1.0, **yellow** at 1.5, **orange** at 2.0, **red** at 3.0 and above. The curve breaks wherever the target is below the horizon or beyond airmass 4.5.
White dot on curve	The target's airmass at the NOW line, with a numeric label

If no target is selected, the plot area shows the message *"Select a target to view its airmass curve"* — the Moon curve, grid and twilight shading are still drawn.

6.2 Scrolling the Time Axis

The plot can be panned horizontally to inspect earlier or later parts of the night:

- **Drag left/right** inside the plot to scroll the time axis. The cursor changes to a grabbing hand while dragging.
- Click the ■ **home button** (bottom-left of the plot) to snap back to the default view centred on the current night.

The NOW line, twilight shading, sunrise/sunset markers and all curves scroll together, so relative timing is always clear. Curves are clipped at the plot border rather than spilling into the axis margins.

6.3 Interactive Cursor

Hover the mouse over the plot to read off values at any time during the night:

- A dashed hairline follows the mouse horizontally
- A time label (e.g. *04:30 UT*) appears along the bottom inside edge of the plot
- A boxed airmass readout, outlined and coloured to match the curve, floats beside the cursor with a dot marking the exact point on the selected target's curve

Moving the mouse out of the plot area dismisses the cursor. Hovering does not interfere with drag-to-pan — a small drag threshold (4 px) distinguishes the two gestures.

6.4 Collapsing & Resizing

Click the **AIRMASS PLOT** header bar to collapse or expand the section; collapsing frees the vertical space for the sky display. Drag the handle just above the header to change the plot's height while it is open. The handle is hidden while the section is collapsed, and no drawing work is done for a collapsed plot.

7 Target File Browser

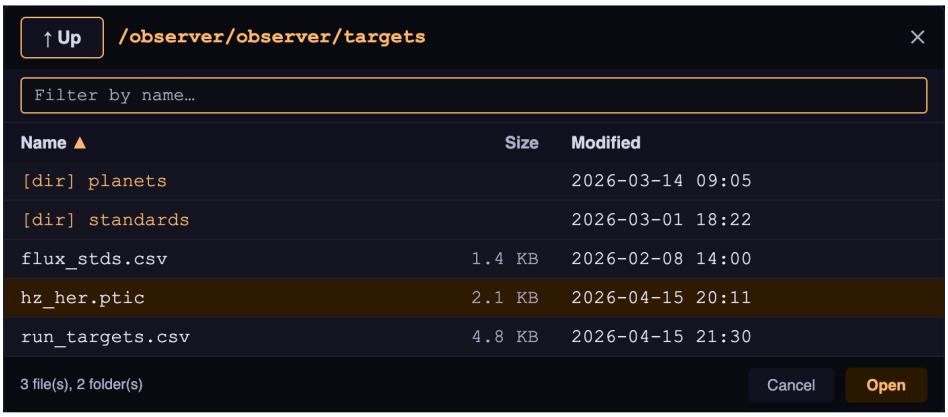


Figure 8 — The file browser opened at `/observer/observer/targets`. Directories are listed first in orange; the selected file is highlighted.

The file browser gives you access to the target-list directory on **observer1.palomar.caltech.edu**. It opens when you click **+ Load Target List** (open mode) or **Save Target List** (save mode) in the Target Lists panel header.

> **Changed in Revision 1.2:** FACSUM now runs on Observer1 itself, so the browser reads and writes the filesystem directly. The former SFTP client — and the username/password dialog it required — has been removed entirely.

Navigating and opening files

Action	How
Enter a directory	Double-click any **[dir]** row
Go up one level	Click ↑ **Up** in the modal header (disabled at the root)
Filter the listing	Type in the *Filter by name…* box (case-insensitive; resets on navigation). The status line shows the number of matches.
Sort the listing	Click a column header (Name, Size, or Modified) to sort; click again to reverse. Directories always sort before files.
Select a file	Click a file row to highlight it
Open a file	Double-click a file row, or click a row and press **Open**
Save to a file	In save mode, navigate to the directory, type or confirm the *Filename*, and press **Save Here**. Clicking an existing file copies its name into the field.
Cancel	Click **Cancel**, the ✕ button, or outside the dialog — nothing is loaded or written

The browser opens at `/observer/observer/targets` and, within a single page load, reopens wherever you last were. Reloading the page — or clicking ✕ **Clear List** — returns it to the default directory.

Access is confined to that directory and its subdirectories: ↑ **Up** stops at the root and any path outside the tree is rejected by the server. The status line at the bottom of the dialog reports the number of files and folders in the current directory, or an error if a directory cannot be read.

FACSUM accepts any plain-text target file in CSV, PTIC or NGPS format (see Appendix A). File extensions (`.csv`, `.ptic`, `.list`, `.txt`, etc.) are not significant — the format is detected from the file content automatically.

A Appendix A — Target File Formats

FACSUM supports three plain-text target list formats. All three accept **comment lines** beginning with `#` or `!`, and **inline comments** after a `!` on any data line. Text after the `!` is treated as notes and is shown in the Notes column of the table.

Lines are parsed *independently*: a file may mix CSV and PTIC lines, and each row remembers which format it came from so that later edits are written back in the same style (A.4).

A.1 CSV Format

Tabular data with consistent column separators (comma, semicolon, tab, space, pipe, or a custom character). Fields are positional and parsed left-to-right:

#	Field	Required?	Format	Example
1	Name	Yes	Any string; quote it if it contains the delimiter	`HZ Her`
2	RA	Yes	`HH:MM:SS.s`, `HH MM SS`, or a decimal value	`18:10:28.1`
3	Dec	Yes	`±DD:MM:SS`, `DD MM SS`, or decimal degrees	`+30:48:22`
4	Epoch	No	J2000, B1950, or a numeric year (defaults to 2000)	`2000`
5	Rate RA	No	Numeric; units set by the Rate Units control	`0.00`
6	Rate Dec	No	Numeric	`0.00`
7	PM RA	No	Numeric, mas/yr — only read if fields 5 and 6 are also numeric	`-1.234`
8	PM Dec	No	Numeric, mas/yr	`3.456`
9+	Notes	No	Any remaining text, or text after `!`	`Flux std`

The four optional numeric fields are consumed greedily and in order: the first non-numeric token stops the sequence, and everything from that point onward becomes Notes. So a line with a rate pair followed by a comment works, but a line with a comment between the rates and the proper motions does not.

Coordinate flexibility

- **Decimal coordinates** are accepted and converted to sexagesimal on load. An RA of 24 or less is read as *hours*; anything larger (up to 360) is read as *degrees*. Dec is always degrees, –90 to +90. Converted cells are shown in italics, and hovering one reveals what was in the file (e.g. *Parsed as 272.617°*).
- **Sexagesimal values must be complete** — `10:08:22.4` parses, `10:08` does not.
- **Space-delimited files** may split coordinates into separate tokens (`HZ_Her 18 10 28.1 +30 48 22 2000`); FACSUM recognises this when there are at least seven tokens and the second is a bare number.
- **Multi-word names** are supported in whitespace-delimited files provided the coordinates are written with colons (`M 57 18:53:35.1 +33:01:45`) — everything before the first sexagesimal token is taken as the name.

Example — comma-separated:

```
# Name,         RA,          Dec,        Epoch
HZ Her,       18:10:28.1,  +30:48:22,  2000
NGC 6543,     17:58:33.4,  +66:37:59,  2000   ! Cat Eye Nebula
Feige 110,    23:19:58.4,  -05:09:56,  2000   ! Flux standard
```

A.2 PTIC Format

Fixed-width format used by the original FACSUM and the Palomar TCS catalog system. The target name occupies the first 17 characters (space-padded); the remainder of the line is split into space-separated tokens.

Columns	Field	Notes
0 – 16	Name (17 chars)	Space-padded to the right
17+	RA	Three tokens: `HH MM SS.s` (required)
	Dec	Three tokens: `±DD MM SS` (required)
	Epoch (optional)	Any later token that looks like an epoch (`2000`, `J2000`, `B1950`). Defaults to 2000 if absent.
	Rates (optional)	A single token of the form `pm=<rateRA>,<rateDec>`. Units are set by the Rate Units control.
	Notes (optional)	Any remaining tokens, plus anything after a `!`

> **PTIC rates need the `pm=` prefix.** Unlike CSV, bare numbers after the epoch are *not* read as rates — they are collected into Notes. Write `pm=12.4,-3.1` to specify tracking rates on a PTIC line.

Example:

```
# Name             RA                Dec            Epoch
HZ Her           18 10 28.1  +30 48 22   2000
NGC 6543         17 58 33.4  +66 37 59   2000  ! Cat Eye Nebula
BD+28 4211       21 51 11.0  +28 51 50   2000
Encke            08 44 12.0  +18 22 05   2000  pm=41.7,-12.3  ! comet
```

> **Auto-detection preference:** When PTIC and a CSV variant parse a file equally well, PTIC wins. If auto-detection picks the wrong format, select the correct one explicitly with the File Type radio buttons.

A.3 NGPS Format

Column-based format used by the **Next Generation Palomar Spectrograph (NGPS)** observing tool. FACSUM recognises it automatically from a header row whose first three columns are `NAME`, `RA`, and `DECL`. Detection takes precedence over CSV and PTIC, and the File Type radio buttons do not apply.

Unlike CSV and PTIC, NGPS coordinates are given in **decimal degrees** (J2000) for both axes. FACSUM converts them for display and for sending to the TCS, and converts back when saving.

Column	Required?	Description
NAME	Yes	Target name

Column	Required?	Description
RA	Yes	Right Ascension in decimal degrees (J2000)
DECL	Yes	Declination in decimal degrees (J2000)
OFFSET_RA	No	RA offset (arcsec); shown in the *Offset RA* column
OFFSET_DEC	No	Dec offset (arcsec); shown in the *Offset Dec* column
COMMENT	No	Free-text comment, shown in the *Comment* column
PRIORITY	No	Integer priority; shown in an extra *Priority* column
MAGNITUDE, **MAGFILTER**	No	Object magnitude and its filter band; shown together in a *Mag / Filter* column
EXPTIME	No	Exposure time (seconds) or `SNR <value>`

Both comma-separated and space-delimited NGPS files are supported. In space-delimited files the `COMMENT` column may contain spaces, so FACSUM parses the trailing structured columns from the end of the line inward and treats whatever is left as the comment. Other NGPS spectrograph columns — `BINSPAT`, `BINSPECT`, `SLITANGLE`, `SLITWIDTH`, `AIRMASS_MAX`, `WRANGE_LOW`, `WRANGE_HIGH`, `CHANNEL` — are recognised for that purpose but are not displayed.

Example — space-delimited:

```
# NGPS target list
NAME            RA         DECL       COMMENT          PRIORITY  MAGNITUDE  MAGFILTER  EXPTIME
HZ_Her          272.6170   30.8061    X-ray binary      1         8.9        V          600
NGC_6543        269.6392   66.6330    Cat Eye Nebula    2        11.2        V          300
```

A.4 How Edits Are Written Back

Editing a cell (§4.4), deleting a row, or adding a target rewrites the raw text — which is what a save actually writes to disk — and each touched line is re-serialised in its own format:

Line format	Written back as
CSV	The same delimiter the line was read with; the name is quoted if it contains that delimiter. Rates are emitted as a pair (defaulting to `0`) if any rate or proper motion is set, followed by the proper-motion pair when present.
PTIC	17-character padded name, six coordinate tokens, epoch, then `pm=<rateRA>,<rateDec>` when rates are set.
NGPS	The file's own column order and delimiter, with RA/Dec converted back to decimal degrees (6 dp).
Added via the form	A comma-delimited CSV line.
Notes (any format)	Appended as a trailing `!comment`.

Untouched lines — including comments, blank lines and lines that failed to parse — are preserved byte-for-byte. One consequence is worth noting for NGPS files: because a structured `COMMENT` column and an inline `!comment` are indistinguishable once parsed, an edited NGPS line settles on the trailing `!comment` form and leaves its COMMENT column empty.

B Appendix B — TCS COORDS Command Reference

When **Load to Telescope** is clicked, FACSUM sends a `COORDS` command from Observer1 to the TCS over a direct TCP connection on port **49200**. The full command format is:

```
COORDS  <RA_h>  <Dec_deg>  <epoch>  <rate_RA>  <rate_Dec>  [mflag]  '<name>'
```

Argument	Type	Description
`RA_h`	Float	Right Ascension in decimal hours (6 decimal places)
`Dec_deg`	Float	Declination in decimal degrees, negative = south (6 decimal places)
`epoch`	Float	Coordinate epoch (e.g. `2000`); any `J`/`B` prefix in the file is stripped
`rate_RA`	Float	RA rate (4 decimal places); `0.0` if rates are not applied
`rate_Dec`	Float	Dec rate (4 decimal places); `0.0` if rates are not applied
`mflag`	Integer (optional)	Motion flag: *absent* = proper motion / no rates; *1* = arcsec/hr on both axes; *2* = RA in sec/hr, Dec in arcsec/hr
`name`	String	Target name in single quotes; any single quotes in the name itself are stripped

The `mflag` argument is emitted only when rates are actually being sent — that is, when *Apply Rates* is checked, *Rate Units* is one of the Non-sid Tracking modes, and at least one rate is non-zero. Otherwise the command carries `0.0 0.0` and no flag.

Examples

Sidereal target, no rates:

```
COORDS 18.174472 30.806167 2000 0.0 0.0 'HZ Her'
```

Non-sidereal target with arcsec/hr rates (mflag 1):

```
COORDS 10.884722 14.235000 2000 12.4000 -3.1000 1 'Mars'
```

Non-sidereal target with sec/hr RA rate (mflag 2):

```
COORDS 10.884722 14.235000 2000 0.8300 -3.1000 2 'Mars'
```

> **Safety:** the FACSUM server accepts *only* `COORDS` commands from the browser; anything else is rejected before a socket is opened, and embedded newlines are stripped. FACSUM cannot slew, offset, focus or otherwise move the telescope.

> **Reminder:** COORDS loads coordinates into the TCS but does not move the telescope. Always issue a separate slew command in the TCS software after sending.

C Appendix C — Files, Logs & Troubleshooting

This appendix is for the observing assistant or support astronomer; ordinary use of FACSUM never requires it. All paths are relative to the FACSUM installation directory on Observer1.

C.1 Runtime files

Path	Purpose
`data/telemetry.json`	The current telemetry frame, rewritten atomically by the daemon about four times a second. Contains the parsed fields plus the raw TCS responses.
`data/daemon.pid`	PID of the running TCS daemon. Its absence means the daemon is not running.
`data/heartbeat`	Touched on every telemetry poll. If it goes stale for 120 s the daemon releases the TCS connection.
`data/allsky_cache.jpg`	Most recent all-sky camera frame, reused for 30 s and served if the camera is unreachable.
`data/daemon_YYYY-MM-DD.log`	Daemon operational log — connects, disconnects, TCS errors. Rotated daily, kept 14 days.
`data/daemon_launch_YYYY-MM-DD.log`	Captures a raw traceback if the daemon dies before its own logging starts. Normally empty.
`data/webserver_YYYY-MM-DD.log`	Output from the PHP built-in web server, written by `start_facsum.sh`. Rotated daily, kept 14 days.

C.2 Managing the daemon by hand

The browser starts and stops the daemon automatically, so this is only needed for diagnosis. On Observer1, in the FACSUM directory:

```
./start_daemon.sh           # start
./start_daemon.sh status    # is it running?
./start_daemon.sh restart   # stop, then start
./start_daemon.sh stop      # stop
```

The daemon can also be run in the foreground — `python3 tcs_daemon.py` — which is the fastest way to watch it negotiate with the TCS.

C.3 Troubleshooting

Symptom	Likely cause and remedy
Browser opens but the page fails to load	The web server on Observer1 did not start. `start_facsum.sh` prints diagnostics in this case — check whether a `php -S` process is running, what holds port 18080, and the tail of today's `webserver_*.log`. Rerunning the script is usually enough.
Header stays *Disconnected*, or reconnection gives up after 5 attempts	The daemon could not be started, most often because `data/` is not writable by the web server. Check permissions, then `./start_daemon.sh status` on Observer1.

Symptom	Likely cause and remedy
Header stuck on *TCS Reconnecting…*	The daemon is alive but cannot reach the TCS. Check today's `daemon_*.log` for connection errors, and confirm the TCS itself is up and accepting connections on port 49200.
Fields freeze but nothing turns red	Expected behaviour during a short TCS outage — the display holds its last values for 12 s before it reports anything (§3.2).
Airmass column shows "—" and the sky is empty	No LST is being received, i.e. there is no live TCS connection. Everything that depends on sidereal time is suppressed until it returns.
All-sky overlay is missing or misaligned	Missing: the camera is unreachable (common in daylight or with the dome closed) — the last good frame is kept. Misaligned: adjust **Cam Align** (§5.3).
A file's rows all show *Failed to Parse*	Auto-detection chose the wrong format. Expand **Parsing Options** and set File Type and Delimiter explicitly, then look at the Edit Text tab to see which lines are still red.
SIMBAD lookup always fails	Observer1 cannot reach `cdsweb.u-strasbg.fr`. Expand *Details* on the error card for the underlying message; coordinates can always be entered by hand.
Two browser tabs are fighting over the connection	Only one tab should be connected. Close the extra tab, then click **Connect** in the one you are keeping.